

JAWAHAR BHARANITHARAN

xEV Diagnostics & HIL Validation Engineer

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PROFESSIONAL SUMMARY

Automotive Diagnostics & HIL Validation Engineer with 3 years of experience specializing in **xEV powertrain** diagnostic documentation development, validation, and automation within high-volume series production programs. Expertise in authoring UDS communication parameters, architecting DDT XML databases, and establishing functional ECU test benches utilizing Vector interfaces, and Renault DDT software. Proven track record of leveraging Python automation and AI tooling to accelerate delivery timelines by 70% and delivering rapid prototype software demos to secure client confidence and unlock new business pipelines.

PROFESSIONAL EXPERIENCE

Renault Nissan Technology & Business Centre India | Chennai

Powertrain Embedded Software Industrialization – xEV Diagnostics Engineer | June 2023 – Present

Powertrain & Diagnostic Engineering

- Developed EV powertrain diagnostic documentations including DTC repair methods, Snapshot DIDs, ECU reprogramming, and system descriptions aligned with vehicle release milestones.
- Created and maintained DDT XML database defining UDS services and parameters (DIDs, RIDs, I/O controls, DTCs) with correct CAN request–response mappings used by the VCI for ECU communication.
- Collaborated on framing technical requirements for diagnostic and powertrain features, ensuring clear specifications for software development and implementation.

Testing, Validation & Cross-Functional Collaboration

- Established ECU validation test bench integrating BMS and ECM ECUs; identified wake-up conditions and enabled stable CAN communication using Vector CANalyzer, CANoe, and Renault DDT.
- Collaborated with system architects, software developers, and validation teams to ensure diagnostic consistency and release readiness.
- Supported after-sales diagnostic clarification requests and resolved inconsistencies in diagnostic specifications.

Software Prototyping, Automation & AI

- Developed a Python-based automation tool for diagnostic document generation, improving productivity by 70%, reducing manual errors and enhancing documentation standardization.
- Formulated an AI agent to automate the generation of software tool documentation, reducing manual writing overhead and expanding technical support accessibility.
- Developed a functional BSW (Basic Software) prototype on a rapid prototyping platform to demonstrate core software capabilities, successfully boosting customer confidence and unlocking new business pipelines.

SKILLS

- **Vehicle Diagnostics & Network Validation:** Vector CANalyzer, CANoe, Renault DDT, ETAS INCA, dSPACE
- **Protocols:** CAN, LIN, UDS
- **Target ECUs:** ECM, HECM, HEVC, BMS, Inverter
- **Programming & Automation:** Python, VBA, Gitlab

EDUCATION

Bachelor of Engineering (B.E) in Mechatronics Engineering CGPA 8.22

Agni College of Technology, Chennai (2019 – 2023)